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Project Synopsis

Project Title:

Design and Development of a Communicable Fault
Passage Indicator for Low-Voltage Distribution
Networks in Bangladesh.

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Background and Rationale

Electrical power supply is essential for Bangladesh's development, industry, public service and living standards. Although distribution generation and distribution infrastructure has improved considerably, faults on the distribution network still cause long periods of outages, especially in rural and complex low voltage networks. Traditional fault identification methods rely on manual checking, trial switching, and field patrols, leading to lengthy outages, higher operating expenses, and inefficient use of maintenance personnel.

In order to do this, this project suggests creating a Communicable Fault Passage Indicator (FPI) specially for low voltage distribution networks in Bangladesh. The proposed system is designed to enable distribution utilities to have relevant, timely, and actionable information about fault occurrence and location, including real-time current monitoring, intelligent fault classification, local visual indication, geographical positioning and remote communication.

Project Objective

The main goal is to design a smart, cost-effective, and communicable fault-detection and fault diagnóstico system that is able to detect and identify abnormal electrical conditions and share the important fault data at the central monitoring platform. The system will be designed to facilitate accelerated fault identification, better maintenance response, minimized field patrols, and better distribution system reliability.

Technical Approach

- Develop an ESP32-based FPI integrating current sensing, fault classification, GPS positioning, and communication.
- Detects and classifies Normal, Peak Load, Theft and fault conditions based on measured current characteristics.
- Implement local LED indication and remote monitoring through a cloud-based dashboard.
- Validate system performance under defined DC and AC operating and fault conditions.

Key Outcomes

- Successfully demonstrated detection and classification of defined normal and abnormal electrical conditions.
- Achieved local fault indication, GPS-based location identification, and remote data communication.
- Enabled live status monitoring and historical fault-data storage through the cloud platform.
- Developed the prototype at an approximate component cost of BDT 13,010.

National and Institutional Significance

- Supports faster fault identification and maintenance response in low-voltage distribution networks.
- Reduces dependence on manual fault-location procedures and prolonged field patrols.
- Contributes to improved distribution-system reliability, operational efficiency, and outage management.
- Provides a foundation for digitally enabled and smart-grid-compatible distribution monitoring in Bangladesh.

Future Development

- Conduct field trials on actual AC distribution networks and long-duration feeder testing by collaborating with DPDC and PDB.
- Develop industrial-grade enclosure, protection, and scalable hardware for utility deployment.
- Integrate LoRaWAN, NB-IoT, or LTE-M for wide-area communication.
- Explore AI/ML-based fault classification and integration with utility SCADA/DMS platforms.